Implement A*Algorithm

Aim :

To Find the shortest path from a source to a destination using the A* algorithm with heuristics.

Explanation:

A* is a graph search algorithm that finds the shortest path using both actual cost and estimated cost (heuristic).

Procedure:

1. Initialize a priority queue with the start node.
2. Maintain g(n) (path cost) and h(n) (heuristic).
3. Expand the lowest f(n) = g(n) + h(n) node.
4. Track path using came_from dictionary.

Python Code:

```
from queue import PriorityQueue

def a_star(graph, start, goal, heuristic):
    open_set = PriorityQueue()
    open_set.put((0, start))
    came_from = {}
    g_score = {node: float('inf') for node in graph}
    g_score[start] = 0

    while not open_set.empty():
        _, current = open_set.get()

        if current == goal:
            path = []
            while current in came_from:
                path.append(current)
                current = came_from[current]
            path.append(start)
            return path[::-1]

        for neighbor, cost in graph[current]:
            tentative_g = g_score[current] + cost
            if tentative_g < g_score[neighbor]:
                came_from[neighbor] = current
                g_score[neighbor] = tentative_g
                f = tentative_g + heuristic[neighbor]
                open_set.put((f, neighbor))
    return None

# Define graph and heuristic
graph = {
    'A': [('B', 1), ('C', 4)],
    'B': [('D', 2), ('E', 5)],
    'C': [('F', 1)],
```

```
    'D': [('G', 2)],
    'E': [('G', 1)],
    'F': [('G', 3)],
    'G': []
}

heuristic = {
    'A': 7, 'B': 6, 'C': 2,
    'D': 1, 'E': 0, 'F': 3, 'G': 0
}

path = a_star(graph, 'A', 'G', heuristic)
print("Shortest path using A*:", path)
```

Output:

```
Shortest path using A*: ['A', 'B', 'D', 'G']
```

Result:

Thus the program of using the MINIMAX algorithm to evaluate the optimal move in a game tree

was executed successfully